Technology Stack

Date	July 9, 2026
Team ID	SWTID-2026-1262
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details:

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	**Frontend / Client-Side**	HTML5, CSS3, Bootstrap 5	Responsive Design & Quick Interactivity**:** Bootstrap enables a modern, scannable, and clean tabbed layout to seamlessly switch between the three core agricultural user scenarios. It ensures the data entry forms fit perfectly on mobile screens for fields or desktops in research labs.

2	**Backend / Server-Side**	Python 3.10+ & Flask Framework	Flask provides minimalist routing and form handling, making it the ideal framework to link standard HTML forms directly to local Python machine learning execution files without heavy enterprise overhead.
3	**Database / Data Storage**	File Persistence with Pickle (.pkl) & Pandas DataFrames	Serialized Model Footprint Storage: Instead of setting up a heavy external relational server database, trained Random Forest classifier arrays are serialized into a single model.pkl layout layer. Pandas handles temporary in-memory feature matrices instantly during model preprocessing.
4	**Cloud / Hosting / Deployment**	Local Python Development Server & Anaconda Base Environments	Optimized Windows/Linux Runtime Pipeline: Standard Anaconda bases provide optimized, pre-compiled wheels for core mathematical libraries like NumPy and Scikit-Learn natively. The local Flask development loop allows instant testing and real-time model evaluation
5	**Version Control & CI/CD**	Git Version Control	Git is utilized locally to keep clean iterations of model architectures train_model.py), project packages (requirements.txt), and backend engines (app.py),preventing environment or script corruption

6	**Third-Party APIs / Other Tools**	Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn	Data-Driven Machine Learning Intelligence: Scikit-Learn handles the mathematical evaluation and execution loops for algorithms like Random Forest, Decision Tree, and KNN. NumPy converts raw crop chemistry vectors into numerical arrays for instant model calculations.